

CERTIFIED BELLMAN OPERATORS FOR COSTLY POLICY REVISION

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We study minimum-cost revision of an installed policy subject to operating guarantees at every state and date, using one common randomized Markov continuation. Backward repair links approximate feasibility to deployment cost. Bellman support envelopes remain valid at operating ties, and a multi-action search theorem separates policy resolution from arithmetic precision. For general Borel dynamics, a repair-density theorem gives an upper approximation without smoothing controlled atoms. Frozen finite experiments compare two complete search constructions with SCIP. Their best independently certified intersection meets an absolute 10^{-3} target in five of sixteen cases; SCIP reports eight numerical target hits, with its lower bounds not independently certified. Earlier root tests met no targets in eight cases; two returned uninformative fallback intervals. Forty-two further root computations expose allowance regimes and numerical floors. The original thirty positive-cost continuous-state randomized intervals remain open. Exact aggregation, observable fibers, and strict randomized savings retain their separate assumptions and economic content.

KEYWORDS: Dynamic programming, Policy revision, Common Markov policies, Global optimization, Verified computation.

1. INTRODUCTION

An organization revising an installed decision rule faces two different questions. How much operating performance can it sacrifice after a particular state and date? How much does implementing the revised rule cost under the states that the rule itself generates? A revision that looks inexpensive locally can change future occupancy and become expensive globally. A collection of inexpensive restart-specific plans need not be one implementable Markov rule. We study the minimum implementation cost subject to an operating guarantee after every restart, with a single continuation at any shared state and date.

The choice variable is a randomized Markov policy. Randomization is drawn anew conditional on the current state and date; it does not introduce a hidden initial coin that selects a persistent continuation. Implementation costs attach to realized departures from the installed rule. The objective averages these costs under a specified initial law, while the operating inequalities hold at every state, including states with zero initial mass. Neither a deterministic restriction nor independent optimization after each restart preserves this economic problem.

The central contribution is an end-to-end certification argument for this common-policy object. Backward repair converts approximate operating feasibility into exact admissibility with an explicit cost modulus. A covering search over shared action probabilities supplies global lower bounds. Bellman support envelopes and simplex-product identities strengthen those bounds without dividing by an operating-action gap. A finite stopping theorem tracks the diameter of each policy box and the error in directed arithmetic separately. Consequently, exact ties do not invalidate certification or finite-model completeness. The theorem does not imply a favorable number of search nodes.

This separation is consequential near indifference. A strictly positive operating-action gap can make nonreference actions rare and yield a quadratic small-allowance bound for a root relaxation. But that is a fixed-model result. A switching surface can drive the minimum gap to zero as a state representation is refined. We retain the strict-gap theorem and its full constants,

and provide a gap-free stopping modulus as the alternative guarantee. Fixed precision is also not unlimited precision: the frozen implementation uses denominator 2^{44} for node bounds and 2^{40} for deployed probabilities, while a theorem covering every positive requested accuracy must allow the rounding error to decrease.

For continuous states, two distinctions remain essential. Valid Bellman support functions can accommodate action-dependent atomic transitions without treating nearby atoms as close in total variation. In addition, a new repair-density argument approximates every feasible Borel common policy in implementation value using repaired simple proposals. It applies to controlled kernels, not only autonomous observable types. For the original affine maintenance model it identifies a constructive, atom-preserving upper-approximation route. This result is one-sided: it does not prove that a finite set of support prices supplies convergent lower endpoints. The paired convergence results for exact aggregation, reset approximation, and observable fibers therefore retain their explicit structural hypotheses.

The numerical evaluation now includes actual global trees rather than only root-node ablations. Twelve designed environments from maintenance, inventory, and queueing families and four near-tie variants were fixed with source and primitive hashes before execution. Both custom methods retain a covering tree; SCIP receives the same common-policy constraints and strengthening information. At the registered absolute width 10^{-3} , the Bellman-envelope formulation closes four cases, the aggregate-product formulation closes five, and their valid intersection with SCIP's independently evaluated candidate policies closes five. SCIP reports eight numerical target hits. Its native lower bounds are not passed off as independent rational certificates. The stronger custom formulation has narrower final intervals in eleven cases, but loses on each of the three largest environments. These outcomes measure a computational tradeoff, not uniform superiority.

The original continuous-state problem remains part of the scientific target. Its thirty positive-cost randomized intervals are still open, even though twenty-seven feasible lotteries cost less than a lower bound valid for every deterministic feasible rule. The nonlinear two-state experiment likewise supplies informative but wide intervals. Those facts remain visible in the article. All earlier proofs, counterexamples, sensitivity exercises, failed calculations, and stopped-control material are preserved in the research archive, with the previous complete article and supplement reproduced in a historical volume. The current argument does not depend on reading that development history.

1.1. *Relation to the literature*

Constrained Markov decision processes supply the natural framework (Altman, 1999). Occupation measures are particularly useful for expected constraints under a designated initial law. Here a separate occupancy measure for every restart would additionally have to agree on conditional action probabilities at shared state–date pairs. Omitting that compatibility would change the feasible set. Our direct formulation uses one probability variable at each such pair. This is a formulation requirement, not a claim that occupation measures cannot express the problem.

Approximation theory for constrained control studies finite representations, policy recovery, and value convergence (Dufour and Prieto-Rumeau, 2013, Saldi, 2018, Saldi, Linder, and Yüksel, 2018). The particular bridge here is exact backward repair under all-restart inequalities. The repair-density theorem uses a finite family of action kernels to dominate every possible occupancy distribution. Approximation occurs in that measure, while repair restores inequalities at every point, not merely almost everywhere. Its upper approximation must not be confused with a paired global convergence theorem for arbitrary controlled continuous dynamics.

The optimization ingredients are established tools. Product envelopes originate in McCormick (1976); the identity that products sum to their common continuation variable follows from the probability simplex. Best-bound spatial search, dynamic programming, and weak duality are not new. The additional result is a policy-compatible stopping modulus that charges feasibility restoration and arithmetic error. We compare with SCIP rather than excluding a strong general-purpose solver (Hojny et al., 2025). Numerical LP proposals use HiGHS; their associated simplex methods are discussed by Huangfu and Hall (2018). Independent residual corrections follow the distinction between computation and verification emphasized by Rump (2010).

Bellman support functions are related to information-relaxation and dual bounds (Brown, Smith, and Sun, 2010, Brown and Smith, 2014). No finite price portfolio is asserted to be a complete dual. Whole-region inequalities, as in semi-infinite optimization (Hettich and Kortanek, 1993), are required for a continuous-state certificate. A certificate on sampled restarts is not substituted for the original universal constraint.

The economic motivation is organizational adjustment under a performance requirement, rather than learning an unknown operating model. Safe policy improvement and constrained learning solve distinct information problems (Laroche, Trichelair, and Tachet des Combes, 2019, Ding, Zhang, Basar, and Jovanovic, 2020, Ni and Kamgarpour, 2025). Numerical economics also emphasizes model structure and state-space approximation (Judd, 1998, Brumm and Scheidegger, 2017). In contrast to estimated replacement models such as Rust (1987), our environments are designed known-model exercises. They do not identify empirical implementation costs or monetary welfare. The institutional meaning of a lottery and its admissible administration charge are stated explicitly below.

2. THE ECONOMIC OBJECT

Let X be a standard Borel space, $A = \{1, \dots, m\}$ a finite action set, and $t = 0, \dots, T - 1$. The known Borel kernels are P_t^a . Borel operating rewards, implementation costs, and terminal payoff satisfy $|r_t| \leq R$, $0 \leq k_t \leq K$, and $|g| \leq G$. Throughout the central results $0 < \beta < 1$ and $\varepsilon > 0$. Write $S_h = \sum_{s=0}^{h-1} \beta^s$, with $S_0 = 0$.

The operating value and the values of a common policy p obey

$$V_T = g, \quad V_t(x) = \max_a \{r_t(x, a) + \beta P_t^a V_{t+1}(x)\}, \quad (1)$$

$$J_T^p = g, \quad J_t^p = \sum_a p_t(a | x) \{r_t(x, a) + \beta P_t^a J_{t+1}^p(x)\}, \quad (2)$$

$$C_T^p = 0, \quad C_t^p = \sum_a p_t(a | x) \{k_t(x, a) + \beta P_t^a C_{t+1}^p(x)\}. \quad (3)$$

The target is

$$K_R(\nu, \varepsilon) = \inf_{p \in \mathcal{F}(\varepsilon)} \int C_0^p d\nu, \quad \mathcal{F}(\varepsilon) = \{p : J_t^p(x) \geq V_t(x) - \varepsilon \text{ for every } (t, x)\}. \quad (4)$$

All policies in this definition are Borel randomized Markov kernels. The finite action set gives Borel maximizing selectors by choosing the first maximizer. An operating-optimal policy is feasible, so the set is nonempty. We do not require an optimizer to exist in an infinite-dimensional Borel policy space; infimum arguments use approximate minimizers.

A representative implementation specification is $k_t(x, a) = w_t(x) \mathbf{1}\{a \neq a_t^0(x)\}$ for an installed rule a^0 . The relevant cost is departure from this installed rule, not a switch from the last physical action. A physical switching friction would require that previous action in the state. A fixed cost of administering a randomized rule is also a distinct primitive.

2.1. Disadvantage, savings, and common continuation

Choose any bounded Borel potential H_t with $H_T = 0$. Define

$$d_t(x, a) = V_t(x) - r_t(x, a) - \beta P_t^a V_{t+1}(x), \quad (5)$$

$$h_t(x, a) = H_t(x) - k_t(x, a) - \beta P_t^a H_{t+1}(x), \quad (6)$$

$$D_t^p = V_t - J_t^p, \quad S_t^p = H_t - C_t^p.$$

Then $d_t \geq 0$, $D_T^p = S_T^p = 0$, and

$$D_t^p = \sum_a p_t(a | x) \{d_t(x, a) + \beta P_t^a D_{t+1}^p(x)\}, \quad (7)$$

$$S_t^p = \sum_a p_t(a | x) \{h_t(x, a) + \beta P_t^a S_{t+1}^p(x)\}. \quad (8)$$

These are identities for the same policy. In the executed finite formulation H is the implementation value of a first-index operating-optimal reference action a^* . Consequently $d_t(x, a^*) = h_t(x, a^*) = 0$. Eliminating the reference probability leaves bilinear terms between nonreference action probabilities and continuation values. The analytic support results below do not require this particular reference choice, and therefore do not rely on its stability when operating actions tie.

3. COMMON-POLICY CERTIFICATION WITHOUT A STRICT ACTION GAP

Set

$$D_C = K \sum_{s=0}^{T-1} (s+1) \beta^s, \quad (9)$$

$$D_J = \max_{1 \leq h \leq T} \left\{ 2R \sum_{s=0}^{h-1} (s+1) \beta^s + 2Gh\beta^h \right\}. \quad (10)$$

If two policies differ by at most θ in action-distribution total variation at every state and date, their operating values differ by at most $D_J \theta$, and their costs by at most $D_C \theta$. To see this, subtract their Bellman equations. A cost at distance s can be affected by $s+1$ action changes, and a terminal operating payoff by h changes. Using the range, rather than the absolute magnitude, of the action values gives the stated constants. Total variation is one half of the ℓ_1 distance.

THEOREM 1—Backward repair: *For $p \in \mathcal{F}(\varepsilon + \delta)$, $\delta \geq 0$, there exists a Borel common policy $\mathcal{R}_\varepsilon p \in \mathcal{F}(\varepsilon)$ such that $J_t^{\mathcal{R}_\varepsilon p} \geq J_t^p$ at every state and date, and*

$$\sup_{t,x} \text{TV}((\mathcal{R}_\varepsilon p)_t, p_t) \leq \rho_\varepsilon(\delta) := \frac{\delta}{(1-\beta)\varepsilon + \delta}. \quad (11)$$

Its cost exceeds that of p by at most $D_C \rho_\varepsilon(\delta)$. In particular,

$$0 \leq K_R(\nu, \varepsilon) - K_R(\nu, \varepsilon + \delta) \leq D_C \rho_\varepsilon(\delta). \quad (12)$$

PROOF: Fix the repaired continuation backward. At (t, x) put $q_a = r_t(x, a) + \beta P_t^a J_{t+1}^{\mathcal{R}_{\varepsilon^p}}(x)$, $q_p = \sum_a p_t(a | x) q_a$, $q_* = \max_a q_a$, and $b = V_t(x) - \varepsilon$. The induction hypothesis gives $q_p \geq V_t - \varepsilon - \delta$ and $q_* \geq V_t - \beta\varepsilon$. If $q_p \geq b$, retain the original action distribution. Otherwise mix it with a first maximizing action in proportion

$$\alpha = \frac{b - q_p}{q_* - q_p}.$$

The denominator is at least $(1 - \beta)\varepsilon + (b - q_p)$, so $\alpha \leq \rho_{\varepsilon}(\delta)$. The repaired value is b , weakly above both q_p and the original value. Every operation is Borel, and the division is used only with a positive denominator. This constructs one continuation, not a mixture of persistent plans. The perturbation bound gives the cost inequality; apply it to approximate minimizers to obtain (12). *Q.E.D.*

The same formula defines $\mathcal{R}_{\varepsilon}q$ for an arbitrary Borel proposal q , because the repaired continuation always supplies $q_* \geq V_t - \beta\varepsilon$. A small uniform cost bound requires a small violation, but existence of the repair does not. This distinction will be used for continuous states.

3.1. Tie-safe support envelopes

For a fixed real price λ , define $W_{T,\lambda}^- = W_{T,\lambda}^+ = 0$ and

$$W_{t,\lambda}^- = \min_a \{h_t - \lambda d_t + \beta P_t^a W_{t+1,\lambda}^-\}, \quad W_{t,\lambda}^+ = \max_a \{h_t - \lambda d_t + \beta P_t^a W_{t+1,\lambda}^+\}. \quad (13)$$

THEOREM 2—Bellman supports at and near ties: *For every Borel common policy, without a uniqueness or positive-gap assumption,*

$$W_{t,\lambda}^- \leq S_t^p - \lambda D_t^p \leq W_{t,\lambda}^+. \quad (14)$$

Consequently any feasible policy satisfies all inequalities (14) together with $0 \leq D_t^p \leq \varepsilon$. For $\lambda \geq 0$, let

$$Z_{T,\lambda} = \lambda g, \quad Z_{t,\lambda} = \max_a \{\lambda r_t - k_t + \beta P_t^a Z_{t+1,\lambda}\}. \quad (15)$$

Then $\ell_{t,\lambda} := \lambda(V_t - \varepsilon) - Z_{t,\lambda}$ is a lower bound for C_t^p , and it is independent of H . For any finite set $\Lambda \subset [0, \infty)$,

$$\int \max\{0, \max_{\lambda \in \Lambda} \ell_{0,\lambda}(x)\} \nu(dx) \leq K_R(\nu, \varepsilon). \quad (16)$$

PROOF: Subtract λ times (7) from (8). Backward induction and the fact that a convex combination lies between its smallest and largest action value give (14). Direct substitution yields $W_{t,\lambda}^+ = H_t - \lambda V_t + Z_{t,\lambda}$. Thus $C_t^p \geq H_t - W_{t,\lambda}^+ - \lambda\varepsilon = \ell_{t,\lambda}$. Pointwise maxima and nonnegativity preserve the lower inequality before integration. No transition has been approximated in this argument. *Q.E.D.*

For a finite model, each inequality in (14) is linear in D, S . The frozen enhanced formulation uses $\lambda = 0$ and $\lambda = \pm 2^j$, $j \in \{-2, 0, 2, 4, 6, 8, 10, 12\}$. The prices are fixed before optimization. Their finiteness makes them inexpensive cuts, not a claim of exact global duality.

PROPOSITION 1—Affine cones for approximate ties: Suppose finite constants $\lambda_- \leq \lambda_+$ and nonnegative numbers e_t satisfy

$$\lambda_- d_t(x, a) - e_t \leq h_t(x, a) \leq \lambda_+ d_t(x, a) + e_t \quad \text{for every } (x, a).$$

With $B_T = 0$ and $B_t = e_t + \beta B_{t+1}$,

$$\lambda_- D_t^p - B_t \leq S_t^p \leq \lambda_+ D_t^p + B_t \quad (17)$$

for every policy. In particular the result permits $d_t = 0$ and nonzero h_t . If, instead, only actions with $d_t \geq \tau > 0$ are classified as separated, every feasible policy assigns their total probability at most ε/τ .

PROOF: Subtract $\lambda_+ D$ from S in the joint Bellman recursion and bound the stage term by e_t . Induction gives the upper inequality. The lower inequality is identical with $-e_t$. For the last claim, nonnegativity in (7) gives $\sum_a p_t(a | x) d_t(x, a) \leq \varepsilon$. Q.E.D.

The affine intercept records the cost of a near-tie group instead of hiding it inside an inverse minimum gap. It need not be small. In particular, an exact operating tie may allow a substantial implementation-cost difference. The frozen solver uses the Bellman supports, not this newly stated affine-cone variant; the latter is an analytic strengthening available for subsequent implementations.

3.2. The strict-gap specialization

The earlier regret-scaled rate remains a useful specialization. With an operating-optimal reference and strictly positive nonreference disadvantages, put

$$\underline{d} = \min_{t, i, a \neq a^*} d_{tia}, \quad \chi = \min\{1, \varepsilon/\underline{d}\},$$

$$\lambda_- = \min\{0, \min h_{tia}/d_{tia}\}, \quad \lambda_+ = \max\{0, \max h_{tia}/d_{tia}\}.$$

Feasible nonreference mass is at most χ , and $\lambda_- D \leq S \leq \lambda_+ D$. With $\Delta P_{tia} = P_i^a - P_i^{a^*}$, define $e_{D, T-1} = 0$ and, for $t < T - 1$,

$$e_{D, t} = \frac{\beta \varepsilon^2}{4} \max_i \sum_{a \neq a^*} \frac{\|\Delta P_{tia}\|_1}{d_{tia}}, \quad E_{D, t} = e_{D, t} + \beta E_{D, t+1}, \quad (18)$$

$$E_{S, t} = (\lambda_+ - \lambda_-) e_{D, t} + \beta E_{S, t+1}, \quad \overline{E}_D = \max_t E_{D, t},$$

with zero terminal errors. An exact optimizing root-LP solution, repaired toward the operating reference, has width bounded by

$$E_{S, 0} + D_C \chi \frac{\overline{E}_D}{(1 - \beta) \varepsilon + \overline{E}_D}. \quad (19)$$

The bound follows from the McCormick product error, at most one quarter of the product of the two interval widths, followed by the rare-departure repair. The full derivation is retained in the historical technical material. At fixed primitives, fixed $\beta < 1$, and $\underline{d} > 0$, (19) is $O(\varepsilon^2)$. A numerical LP optimality loss and directed endpoint errors must be added when an implemented

certificate is compared to that exact-LP statement. A solver status code does not set those losses to zero.

The rate neither covers ties nor remains uniform when $\underline{d}$ vanishes along state refinements. Formula (17) and Theorem 3 below give valid alternatives without declaring the strictly separated theorem uniform. The tables report the actual constants and exclude tied cases from the inverse-gap formula.

3.3. A complete multi-action search and its numerical floor

For finite X , use one coordinate for every nonreference probability at every state and date. A node is a box in these $d = nT(m - 1)$ coordinates, intersected with the local probability simplexes. The reference probability is the remaining mass. The lower corner belongs to every nonempty such row because the sum of its nonreference lower bounds cannot exceed one.

Restricted Bellman minimization of regret and implementation cost over these local simplexes gives, respectively, a necessary operating test and a cost lower bound. These restricted dynamic programs are not the final constrained optimization. They are admissible relaxations of all the common policies in the box. The linear formulation additionally imposes (7)–(8), the support cuts, and product envelopes. In the disaggregated formulation the products also obey

$$\sum_{a \in A} w_{tia,j} = z_{t+1,j}, \quad w_{tia,j} = p_{tia} z_{t+1,j} \quad \text{before relaxation,} \quad (20)$$

for each retained continuation coordinate $z = D, S$. The aggregate formulation instead relaxes $p_{tia}(\Delta P_{tia} z_{t+1})$. These are different relaxations of the same Bellman equations. Disaggregation costs more variables and arithmetic per node.

A floating-point LP supplies only a candidate policy and multipliers. For a minimization LP $\min c'x$ subject to $Ax = b$, $Gx \leq h$, and $l \leq x \leq u$, any equality multipliers y and nonnegative inequality multipliers μ give

$$b'y - h'\mu + \sum_j \min\{(c - A'y + G'\mu)_j l_j, (c - A'y + G'\mu)_j u_j\} \quad (21)$$

as a lower bound. Directed arithmetic encloses each term outward. Zero multipliers remain valid if a proposal is unavailable. A floating-point infeasibility flag alone never removes a box.

The incumbent is the smallest exactly evaluated repaired-policy cost. Every split retains both children. Infeasible leaves require a rational necessary inequality; all other terminal leaves retain their lower endpoint. The global lower endpoint is the minimum of those leaf endpoints and the incumbent. This full cover is part of the proof object.

THEOREM 3—Gap-free finite stopping with explicit precision: *Consider a surviving probability box of maximum nonreference coordinate width w . Suppose each restricted Bellman scalar result is rounded outward with error at most $q = 2^{-b}$ per date, and all other node removals and additional lower bounds are valid. Put*

$$\begin{aligned} \theta(w) &= \min\{1, (m - 1)w\}, & \gamma_b &= 2^{-b} S_T, \\ \mathcal{H}(w, b) &= D_C \theta(w) + \gamma_b + D_C \rho_\varepsilon(D_J \theta(w) + \gamma_b). \end{aligned} \quad (22)$$

The cost of exact backward repair of the lower-corner policy minus the rounded restricted cost lower bound is at most $\mathcal{H}(w, b)$. For every $\eta > 0$, choose b and $w_ > 0$ with $\mathcal{H}(w_*, b) \leq \eta$.*

Best-bound bisection, with a widest-coordinate split at least once every eight levels of a path, then terminates at gap η without node or time caps. One sufficient full-tree depth bound is

$$8d \lceil \log_2(1/w_*) \rceil, \quad (23)$$

where the ceiling is replaced by zero if $w_ \geq 1$. No minimum positive action disadvantage enters these statements.*

PROOF: The detailed argument is in Appendix A. Restricted Bellman minimization attains minimum regret simultaneously at every state with one Markov policy in the box. Its exact minimum regret is at most $\varepsilon + \gamma_b$ in a surviving node. All policies in that box differ by at most $\theta(w)$ in action-distribution total variation. The corner therefore violates operating feasibility by at most $D_J\theta(w) + \gamma_b$. Its cost is at most $D_C\theta(w) + \gamma_b$ above the rounded restricted minimum. Repair proves (22). A small enough box cannot require further splitting when its repaired policy has been considered. Widest-coordinate splits force all widths below w_* by the stated depth, independently of heuristic intervening splits. There are finitely many nodes up to that depth. Q.E.D.

The theorem is uniform over finite models with common bounds $R, K, G, T, m, \beta, \varepsilon$ for the stopping modulus, even when action gaps vanish. It is not uniform in computational dimension: the depth contains d , and the number of tree nodes is exponential in depth. Nor does it give a joint approximation theorem merely by substituting a finer state grid for the original kernel.

The frozen code uses two grids: 2^{-44} for node-bound arithmetic and 2^{-40} for deployed nonreference probabilities. It also uses an operating-reference repair with a rounding guard, rather than the exact first-maximizer repair in Theorem 1. The following additional bound charges this implementation difference.

COROLLARY 1—Quantized operating-reference repair: *Let b_v denote the bound precision and b_p the deployed-probability precision. Define*

$$\begin{aligned} s &= (1 - \beta)\varepsilon, \quad B = \max_{t,i,a} d_{tia} + \beta\varepsilon, \quad \kappa = (m - 1)2^{-b_p}, \quad \mu = \kappa B, \\ \delta &= D_J\theta(w) + \gamma_{b_v}, \quad \Delta = \delta + \mu S_T. \end{aligned} \quad (24)$$

Assume $\mu < s$. At each row, mix the proposal toward the operating reference until its regret is at most $\varepsilon - \kappa \max_a |q_a - q_{a^}|$, where $q_a = d_{tia} + \beta P_i^a \widehat{D}_{t+1}$ uses the already repaired continuation. Round each nonreference probability downward to a multiple of 2^{-b_p} and assign the remaining mass to the reference. This is the frozen repair rule. Its repaired lower-corner cost minus the rounded node cost lower bound is at most*

$$\mathcal{H}_Q(w, b_v, b_p) = D_C\theta(w) + \gamma_{b_v} + D_C \min \left\{ 1, \kappa + \frac{\Delta}{s - \mu + \Delta} \right\}. \quad (25)$$

The finite-depth conclusion of Theorem 3 holds with $\mathcal{H}_Q$ in place of $\mathcal{H}$. Arbitrary positive target accuracy requires choosing both precisions sufficiently large; fixing either grid need not give a vanishing sufficient floor.

PROOF: Appendix A bounds the accumulated regret increase due to rounding by μS_T , the mixing fraction by $\Delta/(s - \mu + \Delta)$, and the extra row total variation by κ . Apply the implementation-cost modulus to the repaired lower corner. The reference slack s is positive without any strict action gap. Q.E.D.

The frozen pair $(b_v, b_p) = (44, 40)$ gives sufficient floors below 10^{-3} on all sixteen recorded models; the largest is about 3.715×10^{-4} for Q3. The corresponding exact-repair floor is smaller and is not substituted for this implemented-repair bound. The supplementary tables give both versions and their sufficient widths. Actual certificate endpoints remain valid independently of these conservative bounds. The pre-existing fixed-precision sources and measurements are unchanged; no adaptive-precision experiment is claimed.

At $\varepsilon = 0$, a different exact argument applies. Because all terms in (7) are nonnegative, every feasible policy uses only operating-maximizing actions at every state and date. Conversely those actions give zero regret by backward induction. Minimizing implementation cost by Bellman recursion within that action set therefore solves the zero-allowance problem, including ties. This statement does not require taking the singular limit of (11). The retained finite-horizon allowance-ladder extension likewise has its own assumptions when $\beta = 1$.

4. CONTROLLED ATOMIC STATES AND APPROXIMATION

A continuously indexed state is economically relevant when it affects operating rewards, implementation costs, or action-dependent transitions. It cannot be dismissed as a nuisance coordinate merely because a separate aggregation example has that property. The following results retain Borel common policies and allow controlled transitions. They distinguish an approximation of an upper policy from a certificate covering the unrestricted optimum.

4.1. Repair density for unrestricted Borel policies

Define finite, not necessarily probability, measures

$$\mu_0 = \nu, \quad \mu_{t+1}(B) = \sum_{a \in A} \int P_t^a(x, B) \mu_t(dx). \quad (26)$$

Their masses are m^t . Every state distribution induced from ν by any admissible or inadmissible Markov policy is dominated by μ_t . Let Π_N be increasing finite Borel partitions generating the Borel sigma field of X . Such partitions exist on a standard Borel space. Let $\mathcal{Q}$ be the countable class of all rational-probability policies that are constant on some Π_N at each date.

THEOREM 4—Repair-complete upper approximation: *Under the bounded Borel assumptions of Section 2,*

$$\inf_{q \in \mathcal{Q}} \int C_0^{\mathcal{R}_\varepsilon q} d\nu = K_R(\nu, \varepsilon). \quad (27)$$

If $(q_j)_{j \geq 1}$ enumerates $\mathcal{Q}$, the exact feasible upper values $U_N = \min_{j \leq N} \int C_0^{\mathcal{R}_\varepsilon q_j} d\nu$ decrease to the unrestricted Borel-policy infimum. No exogeneity, recoverable type label, positive action gap, continuity of the policy, or smoothing of an atomic kernel is assumed. This is an upper-approximation theorem; it supplies neither a finite N error rate nor a matching global lower sequence.

PROOF: For any feasible p , conditional expectations of its probabilities on Π_N converge in $L_1(\mu_t)$ to p_t . Rational simplex approximations preserve that convergence. Write the resulting simple proposal as q^N . A backward projection argument shows that the operating value of $\mathcal{R}_\varepsilon q^N$ approaches J^p in $L_1(\mu_t)$: its scalar value is the maximum of the proposal value against the repaired continuation and $V_t - \varepsilon$, and J_t^p is above that threshold. The repair probability is

bounded by the value discrepancy divided by $(1 - \beta)\varepsilon$. Therefore repaired action distributions approach p in the same measures. A performance-difference identity and occupancy domination then imply convergence of initial implementation costs. Appendix B supplies the complete quantitative induction, including measure-zero states. Every repaired policy is globally feasible, so the reverse inequality in (27) is immediate. Approximate feasible minimizers finish the argument. *Q.E.D.*

The result does not restrict the target policy class to a sieve. Only proposals are simple; their repairs need not be. All-state feasibility is restored by pointwise Bellman construction, including states excluded from every μ_t -integral. Dominating measures can be expensive: their total masses grow as m^t . The theorem therefore removes an expressivity obstruction without claiming a practical dimension-free rate.

For the original maintenance primitives in Section 6, rational interval partitions give a concrete representational consequence. With rational piecewise-affine rewards and finitely many affine successor maps, V is piecewise affine on a finite partition at every finite horizon. If q is rational piecewise constant, the repaired operating recursion is

$$J_t^{\mathcal{R}^q} = \max \left\{ \sum_a q_t(a | x) (r_t + \beta P_t^a J_{t+1}^{\mathcal{R}^q}), V_t - \varepsilon \right\}. \quad (28)$$

It remains piecewise affine. First-index maximizing action selection and the repair fraction are piecewise rational, and the fraction's denominator on its active region is at least $(1 - \beta)\varepsilon$. For installed rules with finitely certified regions and piecewise-affine costs, implementation values are piecewise rational. This representation condition must be checked for a compiled incumbent; it is not inferred for an arbitrary floating-point neural network. Exact breakpoint refinement and convergent outward integration under the uniform initial law can consequently certify each upper value. Jumps and isolated breakpoints are assigned explicitly; an integral ignoring a point is not an excuse to omit its operating constraint. This establishes a constructive upper route for the original controlled atoms. No new optimization of the thirty original positive-cost rows is reported in this revision.

4.2. Support witnesses that preserve moving atoms

Theorem 2 gives global lower bounds even when transitions interact across continuous labels. Suppose certified functions $\underline{V}_t \leq V_t$ and $\bar{Z}_{t,\lambda} \geq Z_{t,\lambda}$ are available. Then

$$\int \max\{0, \max_{\lambda \in \Lambda} [\lambda(\underline{V}_0 - \varepsilon) - \bar{Z}_{0,\lambda}]\} d\nu \leq K_R(\nu, \varepsilon) \quad (29)$$

for finite nonnegative prices. A terminal inequality and a Bellman supersolution for Z establish its upper witness by induction. In a finitely atomic model the operator is evaluated as the original finite weighted sum at its exact successor maps. No weak-to-total-variation substitution is used.

These bounds also have a gap-free primitive stability statement. Suppose corresponding models satisfy $\|V_t - \tilde{V}_t\|_\infty \leq a_t$ and $\|Z_{t,\lambda} - \tilde{Z}_{t,\lambda}\|_\infty \leq z_{t,\lambda}$. Then the support lower functions differ by at most $\lambda a_t + z_{t,\lambda}$ at the same allowance. The bound does not involve a reference selector. For example, with uniform reward error e_r , cost error e_k , terminal error e_g , and transition total variation at most α_t , valid backward errors are

$$\begin{aligned} a_T &= e_g, & a_t &= e_r + \beta a_{t+1} + 2\beta \alpha_t M_{t+1}^V, \\ z_{T,\lambda} &= \lambda e_g, & z_{t,\lambda} &= \lambda e_r + e_k + \beta z_{t+1,\lambda} + 2\beta \alpha_t M_{t+1,\lambda}^Z, \end{aligned} \quad (30)$$

where M^V, M^Z bound absolute values in the comparison model. The factor two follows from the total-variation convention. More generally the last terms may be replaced by verified expectation errors on the particular comparison functions. For controlled affine atoms and Lipschitz comparison functions, those errors can be bounded by weighted displacements of the atoms times the Lipschitz constant. That is a test-function error estimate, not a bound on total variation between distinct Dirac masses. Discontinuities require one-sided cell bounds or explicit breakpoint treatment.

An approximate operating solution must also be charged on the upper side. The retained witness-driven repair theorem uses globally valid operating enclosures and their Bellman residuals, rather than treating an approximate V as exact. The frozen finite experiments use exact rational operating dynamic programming, so they do not test a difficult operating-oracle construction. The new witness formulas state what an application must verify; they are not reported as a completed high-dimensional oracle experiment.

4.3. Where paired continuum convergence is available

The earlier exact aggregation and reset-transfer results are preserved in full. Exact exogenous aggregation requires both pointwise aggregate transition identities and a compatible family of within-cell measures. These hypotheses can hold for atomic transitions. The resulting cost problem is an ordinary finite linear program, and the continuous nuisance coordinates of its service examples do not affect rewards, implementation costs, or regime transitions. This is an exactness test, not evidence of solving an economically coupled high-dimensional continuous model.

Observable fibers give a different paired route. Suppose the state is (i, z) and the next state is $(j, \Phi_t(z))$ with action-dependent regime probability $P_t^a(i, j; z)$. Each Φ_t is an autonomous Borel bijection with Borel inverse. With $F_t = \Phi_{t-1} \circ \dots \circ \Phi_0$, the label $\zeta = F_t^{-1}(z)$ is observable. If the initial law is $\nu(di, d\zeta) = q_i(\zeta)\xi(d\zeta)$ for any Borel probability vector $q(\zeta)$, then

$$K_R(\nu, \varepsilon) = \int v(\zeta, q(\zeta), \varepsilon) \xi(d\zeta), \quad (31)$$

where v is the finite common-policy optimum in that fiber. Independence of initial regime and label is unnecessary. Appendix C gives the full Borel finite-selection construction and the error for approximating the conditional initial law.

Refining a fiber partition alone cannot eliminate residual finite optimization gaps. Both the primitive variation and the finite global intervals must be refined. Nor can (31) apply to the original condition process merely because its shocks are discrete: actions move its continuous condition and do not preserve an autonomous observable label. The repair-density result and the atomic support witnesses address that class directly, but a convergent pairing of their two sides is not established here.

5. FROZEN GLOBAL-SEARCH EVIDENCE

5.1. Design, information, and computational accounting

The global experiment consists of twelve designed cases from three families and four variants of a separate near-tie construction. Within each primary family, (n, T, m) takes values $(4, 4, 3)$, $(8, 8, 3)$, $(8, 16, 4)$, and $(16, 32, 4)$. The corresponding (β, ε) pairs are $(0.9, 0.01)$, $(0.95, 0.01)$, $(0.95, 0.005)$, and $(0.95, 0.001)$. Every method receives identical rational rewards, costs, controlled transition probabilities, initial law, and restart inequalities. The inventory and queueing

primitives use discrete demand or arrival shocks; the maintenance family varies controlled regime transitions. The exact generators are specified in the supplement.

The final source freeze precedes these executions and includes eleven scientific source hashes and sixteen primitive hashes. The registered allowance is 120 seconds and 2,047 nodes per method, with 15 seconds per node LP and at most five seconds for a local root proposal. A budget check occurs between proof-producing node operations: an iteration can exceed the nominal wall cap while completing a valid node or its sibling. Measured time, rather than the nominal cap, is reported. Exact checking time is charged separately. Every method was run with one numerical thread; the environment records Python 3.13.15, NumPy 2.3.5, SciPy 1.17.0, and PySCIPOpt 6.2.1 on a four-CPU Linux runner. Paired methods in a case use the same runner. Hardware equivalence is not inferred across separately scheduled cases.

Method B uses Bellman supports, disaggregated products, and simplex-product consistency. Method A uses the aggregate-product relaxation. Both use the same common-policy repair and covering-tree contract. A local nonlinear optimizer can propose a root candidate, but its output is quantized and repaired before it affects an upper endpoint. Every node also evaluates its lower-corner policy. Thus soundness and theoretical completeness do not depend on success of the local optimizer or on a favorable LP proposal.

SCIP uses the common-policy bilinear equations, the same root domains, Bellman supports, and strict-gap cones when applicable. Its objective, feasibility tolerance, random seed, one-thread setting, absolute-gap target, and budgets are recorded. This grants the external solver the same strengthening information rather than withholding the method’s cuts. Its native lower endpoint is a floating-point solver assertion; its proposed policy is independently evaluated after rational repair. No claim is made that SCIP cannot provide other verification modes. The comparison concerns this executed nonlinear adapter.

5.2. Every global interval, including unsuccessful stops

Table I keeps all sixteen cases. B has narrower final intervals in eleven cases, A in four, and one pair is equal. Nevertheless B meets the target in four cases and A in five. Their certified intersection, including the independently evaluated SCIP policy, meets it in five. A narrower root bound or a larger formulation is not enough to predict final target attainment under a fixed budget.

There is now successful closure after branching. On Q1, B reduces its gap below 10^{-3} in 49 nodes; A requires 153 nodes. On M0, A closes in 13 nodes, whereas B already closes at its root. I0 is a zero-cost case, and Q0 is nearly exact at the root; they are not counted as demonstrations of difficult search. Q3, the largest queueing case, closes under A at the root. B explores only seven nodes and leaves a width about 0.0322 there. All three largest cases favor A’s final width. Proof construction and stronger per-node relaxations consume resources that otherwise support further branching.

SCIP reports eight native target hits. It often returns smaller numerical gaps substantially faster than the custom proof-producing search. Its failures, including node limits on inventory and tie cases and time limits on the largest maintenance and inventory cases, are also retained. This evidence does not establish computational superiority of the new cuts or implementation over a modern external global solver. What the custom method adds in the reported comparison is a replayable rational covering certificate. Both the cost of that certificate and the stronger external numerical performance are visible.

TABLE I
ALL FROZEN GLOBAL CASES: ABSOLUTE WIDTHS AT TARGET 10^{-3}

Case	n	T	m	B width	A width	Intersection	Target met
M0	4	4	3	0.000497	0.000567	0.000497	Yes
M1	8	8	3	0.007157	0.0222	0.007157	No
M2	8	16	4	0.008723	0.0255	0.008723	No
M3	16	32	4	0.7286	0.1150	0.1139	No
I0	4	4	3	0	0	0	Yes
I1	8	8	3	0.0179	0.0268	0.0179	No
I2	8	16	4	0.0142	0.0349	0.0142	No
I3	16	32	4	0.0640	0.004768	0.004630	No
Q0	4	4	3	7.607×10^{-12}	6.129×10^{-12}	6.129×10^{-12}	Yes
Q1	8	8	3	0.000980	0.000997	0.000980	Yes
Q2	8	16	4	0.001151	0.004986	0.001151	No
Q3	16	32	4	0.0322	0.000694	0.000660	Yes
T0	4	8	3	0.004995	0.1606	0.004995	No
T1	4	8	3	0.005256	0.2000	0.005256	No
T2	4	8	3	0.003211	0.1678	0.003211	No
T3	4	8	3	0.1345	1.3603	0.1345	No

M: maintenance; I: inventory; Q: queueing; T: near-tie variants. B and A are independently checked whole-tree intervals, not root LP statuses. The intersection uses the largest checked lower endpoint and the smallest checked feasible upper cost, including SCIP’s repaired candidate. It combines separate computations and is not a single 120-second algorithm. Target tests use exact fractions, not rounded displays. Full endpoints, relative widths, work, and stop reasons are in the supplement.

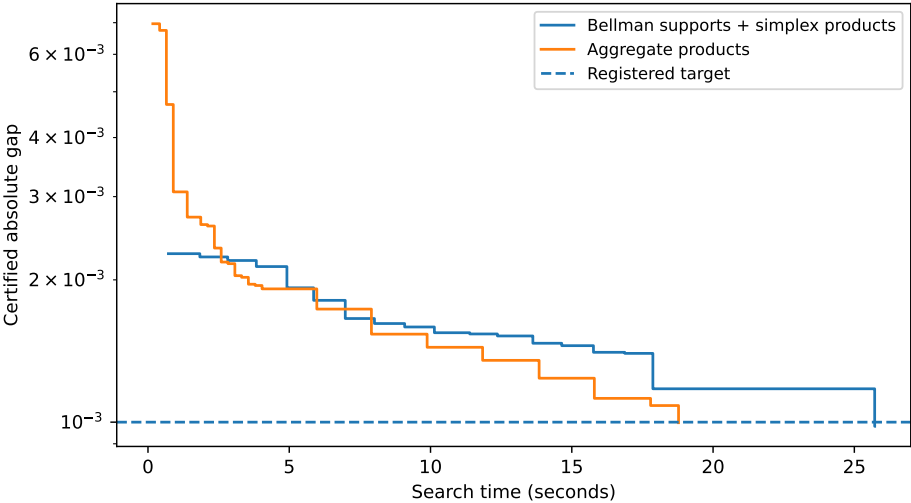


FIGURE 1.—Certified gap along the recorded Q1 searches. Every plotted point is an archived full-cover lower endpoint paired with the current feasible incumbent. Time excludes the final independent checker, whose cost is reported separately. The horizontal line is the registered absolute target.

TABLE II
EXTERNAL GLOBAL BASELINE: NUMERICAL GAPS AND VERIFIED INTERSECTIONS

Case	SCIP stop	Native width	Nodes	Seconds	Checked intersection
M0	gaplimit	0.000497	1	0.11	0.000497
M1	gaplimit	0.000832	11	3.86	0.007157
M2	gaplimit	0.000835	1	12.96	0.008723
M3	timelimit	0.0520	1	120.52	0.1139
I0	optimal	0	0	0.05	0
I1	nodelimit	0.001326	2047	14.26	0.0179
I2	nodelimit	0.003895	2047	72.75	0.0142
I3	timelimit	0.001470	1	120.43	0.004630
Q0	gaplimit	7.000×10^{-9}	1	0.06	6.129×10^{-12}
Q1	gaplimit	0.000243	1	1.35	0.000980
Q2	gaplimit	2.670×10^{-5}	1	4.98	0.001151
Q3	gaplimit	0.000149	1	13.14	0.000660
T0	nodelimit	0.002574	2047	6.55	0.004995
T1	timelimit	0.002388	1359	120.01	0.005256
T2	nodelimit	0.002624	2047	6.17	0.003211
T3	nodelimit	0.1092	2047	10.85	0.1345

Native widths use SCIP’s floating-point primal and dual bounds; they are not independent lower certificates. The final column has the same certified construction as Table I. Native run time includes model construction and candidate repair; independent candidate checking and the two custom lower-bound computations are not hidden inside that time.

5.3. Exact and near operating ties

The tie model has four states, eight periods, three actions, $\beta = 0.95$, and $\varepsilon = 0.01$. The first two operating rewards are 0 and $-\tau$, while the third is $-1/5 - i/40$; terminal operating reward is zero. Actions induce different transitions. Their implementation costs are respectively $1 + i/4$, $7/10 + i/8$, and zero. The four values of τ are 0, 10^{-8} , 10^{-4} , and 10^{-2} . Thus $\tau = 0$ is an exact operating tie with distinct implementation costs and transitions, not two duplicated actions.

B’s final widths in these four cases are approximately 0.004995, 0.005256, 0.003211, and 0.134494. A’s corresponding widths are 0.160574, 0.200032, 0.167849, and 1.360331. All four remain above target, even though the support formulation substantially improves the intervals. The rational validity tests succeed at $\tau = 0$. This demonstrates validity and useful tightening near indifference, not target-level tie closure. The gap-free theorem explains why a positive operating gap is unnecessary for eventual finite-model certification; the observed budgets remain a separate constraint.

5.4. Allowance scaling and the exact-LP distinction

A further frozen calculation uses one $n = 3, T = 8, m = 3$ model from each primary family, seven allowances from 10^{-1} to 10^{-7} , and both earlier root relaxations: 42 root outputs, not 42 independent environments. Each output has its own rational certificate. The complete tables report measured widths, width divided by ε^2 , $\bar{E}_D$, $E_{S,0}$, D_C , χ , the repair fraction, and the sufficient exact-LP bound (19).

The maintenance scaled widths at allowances $10^{-3}, 10^{-4}, 10^{-5}$ are approximately 0.0140484, 1.44063×10^{-4} , and 1.44446×10^{-6} . Their ratios to ε^2 are close to 1.4×10^4 over that range. At 10^{-6} and 10^{-7} the ratios increase, so the finite-precision output is not assigned a universal quadratic exponent. The inventory model contains ties; the strict-gap bound does not apply, and its scaled and unscaled widths coincide. Queueing widths reach roughly 10^{-10} over

TABLE III
ORIGINAL PRIMARY COHORT: OPEN INTERVALS AND STRICT POLICY-CLASS SAVINGS

T	Positive	Median width (%)	Max. (%)	Strict gains	Min. gain	Max. gain
4	10	5.24	14.61	10	0.021898	1.041948
8	10	14.33	32.29	8	0.058701	0.896320
12	10	15.75	34.23	9	0.044193	0.747177

Width is $(U_R - L_R)/U_R$ among positive-cost randomized intervals. A strict gain is $L_D - U_R > 0$, with L_D a deterministic lower bound and U_R a feasible randomized cost. Gain extrema condition on positive gains. The twelve zero-cost configurations are not included in the width statistics. These are retained certificates, not newly optimized rows.

several allowances, making a log–log slope a description of numerical residuals rather than the theoretical rate. The unscaled queueing run at 10^{-6} returns a width about 5.3153 and is retained.

The theorem’s clean bound and the realized interval measure different quantities. The available artifacts do not certify an optimizing primal–dual pair for every numerical root relaxation separately from the deployment repair. Therefore an exact numerical LP-optimality-loss decomposition is marked unavailable, not filled with zeros from an “optimal” status. The full measured rational interval is nevertheless valid. The former four-point scaling calculation and all eight earlier controlled root pairs are retained; none of those eight earlier environments met the 10^{-3} absolute target, and two returned uninformative $[0, U]$ intervals under both relaxations.

6. ECONOMIC INTERPRETATION AND THE RETAINED CONTINUOUS-STATE EVIDENCE

The original maintenance state is $x \in [0, 1]$, with three actions and shock probabilities $3/5, 2/5$. Discounting is $19/20$, the stage operating reward is $x - c_a$ for $c = (0, 3/20, 9/20)$, terminal payoff is $x/2$, and revision weight is $1 + x$. The two successors under each action are

$$(3x/4, 3x/4 + 1/20), \quad (17x/20 + 1/10, 17x/20 + 7/50), \quad (x/10 + 4/5, x/10 + 17/20).$$

These are controlled continuous-state atoms. The primary cohort uses seven installed rules, horizons 4, 8, 12, allowances 0.01, 0.05, and the uniform initial law. The forty-two configurations are a development cohort, not independent observations. All their endpoints and alternative initial-law calculations are preserved.

All thirty positive-cost randomized intervals remain open. Their median relative widths increase from 5.24% at horizon four to 14.33% at horizon eight and 15.75% at horizon twelve. Twenty-seven randomized upper policies nevertheless lie strictly below deterministic lower bounds. These two statements are compatible: a policy-class separation does not identify the randomized optimum. The finite frozen cohort is not a replacement for closing these controlled continuous-state intervals.

6.1. Randomization as an institutional choice

The admissible lottery is a fresh state-dependent action draw at each date, with realized departure charged under k_t . Suppose an organization additionally charges $f \geq 0$ per use of a nondegenerate lottery. For a certified policy p , let

$$B(p) = \mathbb{E}_\nu^p \sum_{t=0}^{T-1} \beta^t \mathbf{1}\{p_t(\cdot \mid X_t) \text{ is nondegenerate}\}.$$

If $s = L_D - U_R > 0$, the same deployed policy remains strictly cheaper than every feasible deterministic rule whenever $fB(p) < s$. A conservative sufficient condition is $fS_T < s$. A one-time fixed governance cost F instead requires $F < s$. This is a break-even implication in the model’s implementation units, not an estimate of actual administrative costs. Such overhead changes the economic objective and must be specified before treating the new optimum as a reported result. Deterministic lower bounds remain a valid comparator because deterministic policies pay no lottery overhead.

For reporting the scale of each separation, the relevant normalized gains are s/L_D and s/U_R , not an operating-welfare percentage. The historical numerical sources retain the signed gains, deterministic bounds, and upper-policy costs. Any percentages computed only from rounded display tables must be labeled approximations rather than exact certificates. No empirical prevalence of feasible organizational randomization is inferred from the designed cohort.

6.2. Other continuous-state classes and the nonlinear experiment

The twelve exact-aggregation service examples retain their tightly closed finite-LP intervals, with atomic but economically ancillary continuous coordinates. The density-transfer exercise retains the failed sufficient tightening at its largest perturbation. The observable-fiber example retains the sixteen-cell interval $[10.074215, 10.184476]$, with width contributions about 0.074982 from finite endpoints and 0.035279 from partition variation. Seventeen endpoint solves represent one parametric family, not seventeen model tests. No newly refined or higher-dimensional fiber computation is claimed.

The nonlinear two-state model has four actions, bilinear successors, and two shocks. At allowance $1/2$, retained relative widths at $(T, N) = (8, 64), (16, 128), (32, 64), (32, 128)$, and $(32, 256)$ are 76.01%, 45.32%, 61.05%, 35.11%, 25.13%. Full independent arithmetic coverage is explicitly marked in the retained tables; in particular the finest $(32, 256)$ object has been independently rebuilt. That audit does not turn a 25.13% interval into a near-exact result. At allowance 0.05, the retained calculations remain lower-only because no matching upper policy is certified. A failed sufficient enclosure is not proof of policy infeasibility. This revision does not invent a per-candidate failure classification when the retained records do not provide one.

The inherited binary finite-state comparison also remains unfavorable to interval Bellman search on every difficult pair: the verified McCormick formulation gives stronger intervals there. The stopped-control derivations, economic sensitivities, deterministic frontiers, and earlier counterexamples remain in the historical volume without being counted as new completed optimization targets.

7. VERIFICATION AND IMPLEMENTATION BOUNDARIES

Three distinct claims require separate evidence. The economic and mathematical reduction must preserve (4). The arithmetic must prove each serialized inequality. The publication must match the checked source and outputs. Hashes address identity, not theorem validity. Finite adversarial tests probe the checker, not the full space of possible bugs.

The standalone tree reader imports neither optimizer nor constructor. It independently rebuilds the model, operating value, reference, supports, product domains, dual residual bounds, policy values, and every split of the global cover. Every feasible upper is checked after every state and date. SCIP candidate checks verify only the feasible policy and its cost, not SCIP’s native global lower bound. The local replication of this revision checks all 32 complete custom trees and all 16 SCIP policies against their frozen model files; the exact reports accompany the manuscript. The 42 root-scaling certificates are separately identified and checked by the earlier standalone root reader.

The frozen transformation audit checked 81 deterministic and 64 mixed policies, with 19,720 support inequalities, including an exact operating tie. Seventeen archived adversarial mutations were rejected. They include forged endpoints, invalid probabilities, dual signs, nonstochastic transitions, altered discount factors, incorrect primitive disadvantages, exchanged action labels, malformed probability/product boxes, forged repair rows, missing leaves or siblings, a coverage gap, and a cyclic tree. These are finite tests, not machine-checked proofs of the general support or stopping theorems. The theorem–code matrix records which theorem is implemented, which assumptions are analytic, and which calculations were actually executed.

The final manuscript and response introduce analytic results beyond the frozen code: the explicit multi-action numerical-floor theorem, affine approximate-tie cones, and repair-density theorem. They are accompanied by proofs and separated from the predeclared numerical claims. The frozen implementations and unfavorable outputs remain unchanged. In particular, an adaptive-precision extension and the atom-preserving upper enumeration are not described as completed numerical experiments.

The new precision contract was also checked with a separate exact-rational implementation on 256 seeded two-state, three-date, three-action boxes, including an exact operating tie. Every surviving box satisfied the ideal-repair stopping inequality and all repaired restart constraints. A separate exact implementation checked the quantized operating-reference repair on those same 256 boxes, including its rounding guard; all also satisfied the two-grid bound. These are two contract checks on the same boxes, not 512 independent test environments. Sixteen additional exact calculations report sufficient precision floors and box widths for the frozen models. These tests check instances of the mathematical contract; the proofs, not the test count, establish the theorems. All forty-two frozen scaling proof objects were independently replayed as well.

8. CONCLUSION

Minimum-cost revision under all-restart operating protection is a common-continuation problem. Backward feasibility repair provides an explicit connection between numerical approximation and admissible deployment. Bellman supports and a multi-action covering search extend this connection to operating ties, while a precision-dependent stopping modulus makes the role of finite arithmetic explicit. The strictly separated small-allowance rate is retained as a sharper conditional result, not a universal numerical law.

For controlled continuous states, the repair-density theorem keeps the unrestricted Borel policy target and permits action-dependent atoms. It supplies a convergent upper approximation; valid support witnesses supply lower bounds without atom smoothing. A matched two-sided convergence rate for the original controlled continuum remains a distinct task. Exact aggregation and observable-fiber identities continue to give paired results under their stated structure.

The complete-search evaluation now records real branching, a modern external baseline, all failed stops, and complete arithmetic covers. Five of sixteen combined rational intervals meet the registered target. The stronger custom relaxation does not dominate the cheaper one at the largest scale, and SCIP often delivers better native numerical performance. Those results identify the value and cost of independently verifiable certification. They do not erase the original thirty open positive-cost randomized intervals or the economic assumptions behind admissible randomization. The policy-class savings and the certification methods are substantive results on their stated objects, not substitutes for unperformed calibration or unsupported claims of universal tractability.

APPENDIX A: PROOF OF THE PRECISION-DEPENDENT SEARCH BOUND

For a node Q , the admissible distributions in each row form a compact simplex intersected with coordinate intervals. If the sum of lower nonreference bounds exceeds one, the row is empty and removal is exact. Otherwise its lower corner is admissible. Given two row distributions p, q in the box,

$$\text{TV}(p, q) = \frac{1}{2} \left(\sum_{a \neq a^*} |p_a - q_a| + \left| \sum_{a \neq a^*} (p_a - q_a) \right| \right) \leq \sum_{a \neq a^*} |p_a - q_a| \leq (m-1)w.$$

The bound is also at most one. It does not depend on which action was used as the reference.

Define $D_{t,Q}^{\min}$ by restricted Bellman minimization of nonnegative disadvantages d on those row simplexes, with zero terminal value. A minimizing action distribution can be selected at each row, and the collection of these distributions is one common Markov policy. Backward induction proves that its regret equals $D_{t,Q}^{\min}$ simultaneously at all state–date pairs. Likewise restricted minimization of k produces $C_{t,Q}^{\min}$. The two minimizing policies can differ: no equality of their choices is asserted or needed.

If each minimum is rounded downward by at most q , the error relative to the exact restricted minimum at a date with h decisions remaining is at most qS_h . This follows from the stochastic row sums and the recursion $e_t \leq q + \beta e_{t+1}$. A surviving node has rounded minimum regret at most ε at every pair, hence $D_{t,Q}^{\min} \leq \varepsilon + \gamma_b$. Let p^c denote its lower corner. The policy perturbation bound gives

$$D_t^{p^c} \leq D_{t,Q}^{\min} + D_J \theta(w) \leq \varepsilon + \gamma_b + D_J \theta(w).$$

If $\underline{c}_Q$ is the initial-law average of the rounded minimum cost, then

$$\int C_0^{p^c} d\nu - \underline{c}_Q \leq D_C \theta(w) + \gamma_b.$$

Repairing p^c and using Theorem 1 gives $U_Q - \underline{c}_Q \leq \mathcal{H}(w, b)$. Raising the lower endpoint by any valid LP residual or inherited parent bound preserves this inequality. Additional valid node removals cannot discard an actually feasible point.

Suppose the incumbent is U and a node’s valid lower endpoint is L_Q . Once its corner has been evaluated, $U \leq U_Q$. If $\mathcal{H}(w, b) \leq \eta$, then $U - L_Q \leq \eta$ and the node need not be refined. Best-bound processing therefore cannot continue splitting such nodes while a positive gap larger than η remains. At least one widest-coordinate split in each block of eight levels reduces all widths below w_* after at most $8d \lceil \log_2(1/w_*) \rceil$ levels: after d widest splits the largest width is no more than half its previous worst-case bound, and intervening splits only help. A full binary tree of that finite depth bounds the number of needed evaluations.

For any fixed positive ε and $\beta < 1$, first choose b so that $\mathcal{H}(0, b) < \eta$, then choose $w_* > 0$ by continuity. Fixed b alone yields a nonzero sufficient numerical floor. The implemented 2^{-44} directed arithmetic is rational and valid, but the statement “every prescribed positive accuracy” is justified only by this precision choice or a precision-refinement schedule. Completeness does not depend on obtaining exact LP optima, because the restricted Bellman lower bound and corner repair suffice.

A.1. The deployed-probability grid and its repair guard

We prove Corollary 1. Use the original lower-corner policy p^c as the comparison, not its ideal repair. The implemented repair works backwards. If future rows satisfy $0 \leq \hat{D}_{t+1} \leq \varepsilon$, the

operating-reference action has $q_{a^*} \leq \beta\varepsilon$, and all action regrets lie in $[0, B]$. Thus its exact row guard $g_{ti} = \kappa \max_a |q_a - q_{a^*}|$ is at most $\mu < s$. Its target $\varepsilon - g_{ti}$ is strictly above q_{a^*} , so the fallback to the reference due to an inadequate target is not needed under this sufficient precision condition.

Mixing changes expected regret to its minimum with that target. Downward rounding of the $m - 1$ free probabilities moves total variation by at most κ and changes expected regret upward by at most g_{ti} . The final regret therefore remains at most ε . It is also at most the proposal regret against the implemented future plus μ . Let $e_t = \sup_i (\hat{D}_t(i) - D_t^{p^c}(i))_+$. Then $e_T = 0$ and

$$e_t \leq \mu + \beta e_{t+1} \leq \mu S_{T-t}.$$

The unmodified corner is feasible at allowance $\varepsilon + \delta$ by the preceding box argument. Consequently the excess of the current proposal regret over its guarded target is at most $\delta + \mu S_T = \Delta$. If that excess is $u > 0$, the mixing fraction is

$$\frac{u}{u + \varepsilon - g_{ti} - q_{a^*}} \leq \frac{u}{u + s - \mu} \leq \frac{\Delta}{\Delta + s - \mu}.$$

The total row variation from p^c to the deployed policy is at most this fraction plus κ , and at most one. Its implementation-cost difference is bounded by D_C times that quantity. Adding the corner's $D_C\theta(w) + \gamma_{b_v}$ distance from the rounded restricted cost minimum proves (25). This proof explicitly accommodates regret increases caused by rounding; it does not assume the implemented rounding step is operating-value monotone.

For any fixed model and positive target, increasing b_p makes $\mu < s$ and sends both μ and κ to zero. Increasing b_v sends γ_{b_v} to zero. Hence $\mathcal{H}_Q(0, b_v, b_p)$ tends to zero, after which the same small-box and widest-split argument supplies a finite stopping depth. The fixed-grid sufficient bounds in the supplement are exact calculations on the serialized models, not claims of an efficient search complexity.

APPENDIX B: QUANTITATIVE REPAIR-DENSITY ARGUMENT

Let $p \in \mathcal{F}(\varepsilon)$, let q be any Borel proposal, and write $\hat{q} = \mathcal{R}_\varepsilon q$. Define

$$a_t = \int \text{TV}(q_t, p_t) d\mu_t, \quad E_t = \int |J_t^{\hat{q}} - J_t^p| d\mu_t, \quad M_t = RS_{T-t} + G\beta^{T-t}, \quad s = (1 - \beta)\varepsilon.$$

We have $E_T = 0$. The action values against any valid continuation are in $[-M_t, M_t]$. Subtracting the proposal Bellman value against the repaired continuation from J_t^p gives a term bounded by $2M_t \text{TV}(q_t, p_t)$ and the continuation term $\beta P_t^{p_t} |J_{t+1}^{\hat{q}} - J_{t+1}^p|$. The scalar repair replaces the former proposal value by its maximum with $V_t - \varepsilon$. Since $J_t^p \geq V_t - \varepsilon$, this projection cannot increase its distance from J_t^p . Moreover, for any nonnegative f ,

$$\int P_t^{p_t} f d\mu_t \leq \sum_a \int P_t^a f d\mu_t = \int f d\mu_{t+1}.$$

Consequently

$$E_t \leq 2M_t a_t + \beta E_{t+1}. \tag{32}$$

The repair fraction at each point is at most the positive deficit divided by s , and that deficit is at most the absolute difference of the proposal value from J_t^p . The triangle inequality therefore

gives

$$\int \text{TV}(\hat{q}_t, p_t) d\mu_t \leq a_t + \frac{2M_t a_t + \beta E_{t+1}}{s}. \quad (33)$$

Both inequalities hold without a uniform approximation of p .

For any policy u , its state laws η_t^u from ν satisfy $\eta_t^u \leq \mu_t$ by induction. The implementation performance-difference identity between $\hat{q}$ and p uses the state law of $\hat{q}$ and the implementation action values of p . Those action values belong to $[0, KS_{T-t}]$. Hence

$$\left| \int (C_0^{\hat{q}} - C_0^p) d\nu \right| \leq \sum_{t=0}^{T-1} \beta^t KS_{T-t} \left(a_t + \frac{2M_t a_t + \beta E_{t+1}}{s} \right). \quad (34)$$

This identity follows directly by telescoping the difference of the two cost Bellman equations; it does not impose an almost-sure operating constraint in place of the pointwise one.

The bounded conditional expectations $\mathbb{E}_{\mu_t}[p_t(a \mid \cdot) \mid \Pi_N]$ converge in $L_1(\mu_t)$ on increasing generating partitions. If a cell has zero measure, choose an arbitrary probability vector there. At each finite partition round all but one probabilities down to multiples of an increasing denominator, assigning the remainder to the last action. The resulting vectors remain probabilities, and their added total-variation error tends uniformly to zero. Because each μ_t is finite, this supplies rational simple proposals q^N with every $a_t \rightarrow 0$.

Equations (32)–(34) imply initial implementation-cost convergence. Meanwhile backward repair enforces $J_t^{\hat{q}^N}(x) \geq V_t(x) - \varepsilon$ for every x , including cells and singletons that have zero dominating measure. For any $\zeta > 0$, choose a feasible p with cost less than $K_R + \zeta$ and then a repaired simple proposal within ζ in cost. Since all members of the countable repaired class are feasible, its infimum equals (4). Enumeration gives the monotone exact upper sequence.

For a numerical use, the Bellman repair and integration must themselves be computable with certified errors tending to zero. This is an additional evaluation requirement, not implied merely by Borel measurability. The finite-horizon rational piecewise-affine maintenance specification has a finite piecewise-rational construction as described in the main text. Its number of pieces can grow rapidly. Neither the present proof nor the frozen finite experiments supply a uniform complexity bound for that construction or a two-sided convergence rate for its global constrained optimum.

APPENDIX C: CONDITIONAL INITIAL LAWS IN OBSERVABLE FIBERS

For each label ζ , let the finite fiber use the primitives evaluated at $F_t(\zeta)$ and initial law $q(\zeta)$. A Borel global policy restricts to a feasible common policy in every fiber. Its cost is therefore at least $v(\zeta, q(\zeta), \varepsilon)$ before integration.

For the reverse inequality, fix a denominator M . There are finitely many simplex-lattice policy arrays for an n -state, T -date fiber. Their Bellman values are Borel in ζ , as finite sums and compositions of the Borel primitives. Their initial costs, weighted by the Borel vector $q(\zeta)$, are Borel as well. Let $\theta_M = \min\{1, (m-1)/M\}$. Rounding an arbitrary feasible policy shows that the lattice feasible set at allowance $\varepsilon + D_J \theta_M$ is nonempty. Its first cost minimizer under a fixed ordering is Borel: every selection event is a finite intersection of Borel comparison sets, including the feasibility comparisons at all finite state–date pairs.

Repair that selected array by Theorem 1. Its dependence on ζ remains Borel, including at ties, because first-index finite maximizers and division on a strictly positive denominator are Borel. Its feasible cost is at most

$$v(\zeta, q(\zeta), \varepsilon) + D_C \{\theta_M + \rho_\varepsilon(D_J \theta_M)\},$$

uniformly in ζ and in $q(\zeta)$. This also proves measurability of v through the corresponding lattice lower and upper endpoints. Lifting the selected array at date t using $F_t^{-1}(z)$ yields one Borel Markov policy, with no remembered latent label. Integrating and letting $M \rightarrow \infty$ proves (31).

For cell approximation, let a representative use q_c while the true conditional initial law obeys $\sup_{\zeta \in B_c} \text{TV}(q(\zeta), q_c) \leq \delta_{q,c}$. Every feasible policy has statewise initial implementation values in $[0, KS_T]$. Changing its initial regime law therefore changes cost by at most $KS_T \delta_{q,c}$. The same bound applies to the infimum because its all-restart feasibility set is independent of the initial law. It is added to each cell’s existing primitive/solver cost error. Alternatively one may integrate the representative policy’s statewise costs against the exact varying conditional law. Independence between regime and label is a convenience of the earlier example, not an assumption needed for disintegration.

APPENDIX: REFERENCES

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